

FAST NATIONAL UNIVERSITY

PROCESS MINING AND SIMULATION
SE4009

Project

Process Mining: CDA Bus Route Analysis



Instructor:

Dr. Isma ul Hassan

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Submission Guidelines

Late submissions will NOT be accepted under any circumstances.

Submission Format

Submit a single .zip archive named:

`member1-id_member2-id.zip` (extend for 3–4 members)

The archive must contain the following folders and files:

- **report/** — A PDF report documenting all tasks, outputs, analysis, and interpretations. The report must include screenshots of outputs, process maps, trace logs, GUI screenshots, bottleneck annotations, and per-member route maps.
- **code/** — All Python scripts and notebooks used to produce results. Code must be clearly commented and structured by task.
- **data/** — The extracted `routes.csv` and any intermediate event log files.

Overview

This project applies process mining techniques to real-world public transport data from the **CDA (Capital Development Authority) bus network** of Islamabad (cda.gov.pk/cdaTransitMap). You will extract structured route data from the published PDF schedules, construct event logs with trace-level granularity, discover and visualise process models across all routes, and build an interactive GUI enriched with timing and bottleneck analytics. An agentic AI component provides natural-language trip-planning over the discovered routes, and a personal mapping task grounds each team member in the network topology.

Mark Distribution

Task	Description	Marks
Task 1	Data Extraction & CSV Generation	10
Task 2	Trace Log Construction	15
Task 3	Process Discovery & Interactive GUI	35
Task 4	Performance & Bottleneck Analytics	10
Task 5	Agentic AI Trip Planner	15
Task 6	Personal Route Map — Bonus (per member)	10
Total		85 + 10 Bonus

1 Task 1 — Data Extraction & CSV Generation

10 Marks

Objective

Inspect the CDA bus route PDFs available at cda.gov.pk/cdaTransitMap and extract all route data into a structured `routes.csv` file.

Requirements

- Download and parse each route PDF from the CDA transit map portal.
- The output file must contain the relevant columns as described in the PDF schedules (e.g. route identifier, stop names, timing).

Marking Criteria — Task 1

Correct extraction of all stops from all PDFs	5
Proper CSV structure and column naming	5

2 Task 2 — Trace Log Construction

15 Marks

Objective

Construct a process mining event log from the dataset by exporting it as a **.xes** file that follows the XES standard, and validate it using a process mining tool.

Requirements

- Generate a single merged **.xes** file representing the dataset as an event log.
- Ensure the file strictly adheres to the **XES standard**, including correct structure, attribute types, timestamps (ISO 8601 with timezone), and required metadata (e.g. **xes.version="1.0"**).
- Import the **.xes** file into a process mining tool (e.g. ProM, PM4Py, Apromore, or Celonis).
- Demonstrate successful loading and analysis of the log by including a screenshot showing trace/event statistics or model discovery results.

Marking Criteria — Task 2

Valid and XES-compliant .xes file	5
Successful tool validation, usage, and evidence	10

3 Task 3 — Process Discovery & Interactive GUI

35 Marks

Objective

Discover a process map from the event logs and present it in an interactive graphical user interface (GUI) that supports filtering and rich annotations.

Requirements

3a — Process Map Generation (10 marks)

- Use a process discovery algorithm on the merged event log.
- The map must show all routes connected, with nodes representing stops and directed edges representing transitions.
- Edge labels must display the transition duration between consecutive stops.

3b — Route Filter (10 marks)

- Provide a dropdown or selector to filter the process map by **route_id**.
- Selecting **FR-01** must display only the sub-graph for that route.
- Selecting **All Routes** must display the full connected map.

3c — Transition Duration on Edges (10 marks)

- Each directed edge must display the transition time between consecutive stops.
- Format: **PAEC → IMC = 1min45sec** (derived from 10:15:05 – 10:13:20).
- This is the **only required annotation on edges**. Additional analytics (throughput, bottleneck info, etc.) may be displayed elsewhere in the GUI (e.g. a side panel, tooltip, or summary table) — not necessarily on the edges.

3d — Case Frequency Display (5 marks)

- Display the number of trips (cases) that traversed each transition.
- This may be shown on edges, in tooltips, or in a separate panel — any clear presentation is acceptable.

Marking Criteria — Task 3

Correct process map from event log	10
Working route filter with correct sub-graphs	10
Transition duration labels on edges	10
Case frequency display (any location in GUI)	5

4 Task 4 — Performance & Bottleneck Analytics

10 Marks

Objective

Enrich the GUI with statistical timing information and automatically highlight bottlenecks (delays) in the network. Analytics may be displayed in any part of the GUI — side panels, summary tables, tooltips, or info boxes are all acceptable.

4a — Throughput Time per Trip (5 marks)

- Compute the total end-to-end duration for each case: $T_{\text{total}} = t_{\text{last departure}} - t_{\text{first arrival}}$.
- Show average, minimum, and maximum trip duration in the GUI.
- Example: 10:13:00 → 10:45:00 ≈ 32 minutes.

4b — Bottleneck Detection & Highlighting (5 marks)

- Identify edges whose average transition time exceeds a configurable threshold.
- Highlight bottleneck edges visually in the process map.
- Provide a bottleneck summary listing the top-3 slowest transitions anywhere in the GUI (panel, table, or tooltip).
- Example bottleneck: OGTI → CDA if it is consistently slower than other edges.

Marking Criteria — Task 4

Throughput time per trip displayed (avg/min/max)	5
Bottleneck detection and visual highlighting	5

5 Task 5 — Agentic AI Trip Planner

15 Marks

Objective

Modern process mining platforms such as **Celonis** demonstrate how agentic AI can be embedded directly into a business process context, allowing users to query, navigate, and act on live process data through natural language. In this task, you will replicate this idea at the route level: integrate a conversational AI agent into your GUI that answers natural-language trip-planning queries grounded in the discovered CDA route data. The choice of model, API, or framework is entirely up to you.

Requirements

- Integrate an AI agent that has access to the extracted **routes.csv** and the discovered process map.
- The agent must be capable of handling queries such as:
 - “I have to travel from Khanna Pul to NUST Metro Station — what are my options?”
 - “Which route goes through Faizabad?”
 - “What time does the last bus leave from H-9?”
 - “How long does it take to get from Khanna Pul to Faizabad?”
 - “Do any routes connect G-9 Markaz to F-10 Markaz?”
- The agent should return: available route(s) and required transfers, estimated travel time based on discovered transition durations, and the next available departure time.
- Responses must be grounded in the route data — the agent must not hallucinate stops.
- A chat interface panel must be embedded in the main GUI.

Example Interaction

User: I have to travel from Khanna Pul to NUST Metro Station. What are my options?

Agent: Based on the current schedule, Route **FR-02** stops at *Khanna Pul* (dep. 10:20) and reaches *NUST Metro Station* in approximately **18 minutes** (avg). Next departure: **10:35**.

Marking Criteria — Task 5

Correct route lookup and path finding	5
Natural-language response quality	5
Integration in GUI chat panel	3
Grounding in route data (no hallucination)	2

6 Task 6 — Personal Route Map (Per Member) — *Bonus*

10 Marks

Objective

Each team member individually maps at least one route connecting their home address (or the nearest bus stop to their home) to **FAST University**.

Requirements

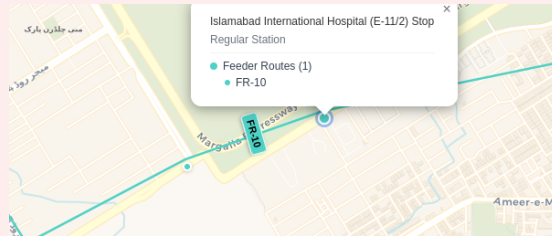
- Each member must identify the nearest CDA bus stop to their home address.
- Construct a trace/path from FAST University to that stop using the available routes.
- Visualise the personal route on a map.
- Include in the report: member name, home area, nearest stop, route(s) used, and estimated travel time.

Example — Member Route

Member: Ali Raza **Home Area:** Satellite Town, Rawalpindi
Nearest Stop: Faizabad Interchange
Route: Faizabad $\xrightarrow{\text{FR-07}}$ H-9 $\xrightarrow{\text{FR-03}}$ FAST
Est. Time: $\approx$ 28 min

This task is bonus (extra credit) and is NOT part of the 85-mark total.

Half-hearted or incomplete attempts will **not** be awarded marks. A good-faith submission looks like the example below.



Example: acceptable personal route map visualisation

Marking Criteria — Task 6

Map produced for every team member	4
Correct stop identification and route tracing	4
Visualisation quality and report documentation	2

Mark Summary

Complete Mark Distribution		
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1	Data Extraction & CSV Generation	10
2	Trace Log Construction	15
3	Process Discovery & Interactive GUI	35
4	Performance & Bottleneck Analytics	10
5	Agentic AI Trip Planner	15
Subtotal		85
6	Personal Route Map — Bonus (per member)	+10
Maximum Possible		85 + 10